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# Purpose: This notebook performs the tasks outlined in 4.2 Assignment: R/Python Refresher. ALS Data will be loaded into the notebook and processed for silo scores & clustering.

Remove any data that is not relevant to the patient's ALS condition. Apply a standard scalar to the data. Create a plot of the cluster silhouette score versus the number of clusters in a K-means cluster. Use the plot created in (3) to choose an optimal number of clusters for K-means. Justify your choice. Fit a K-means model to the data with the optimal number of clusters chosen in part (4). Fit a PCA transformation with two features to the scaled data. Make a scatterplot the PCA transformed data coloring each point by its cluster value. Summarize your results and make a conclusion.

Note! We will remove the irrelevant features at the end to show the difference in silo scores with and without noisy data.

In [2]:
```
# Imports
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import f_oneway
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.decomposition import PCA

import matplotlib.pyplot as plt
# Suppress warnings for cleaner output
import warnings
warnings.filterwarnings("ignore")

# Ensure that all code cells are fully visible without scrolling
from IPython.core.display import display, HTML
```

```
display(HTML('''
<style>
div.CodeMirror pre {
    white-space: pre !important;
}
</style>
'''))
```

In [3]:
```
# Read data into a DataFrame
df = pd.read_csv('als_data.csv')
```

1. Remove any data that is not relevant to the patient's ALS condition.

In [4]:
```
# Drop irrelevant columns
df = df.drop(['SubjectID', 'ID',], axis=1)
```

2. Apply a standard scalar to the data.

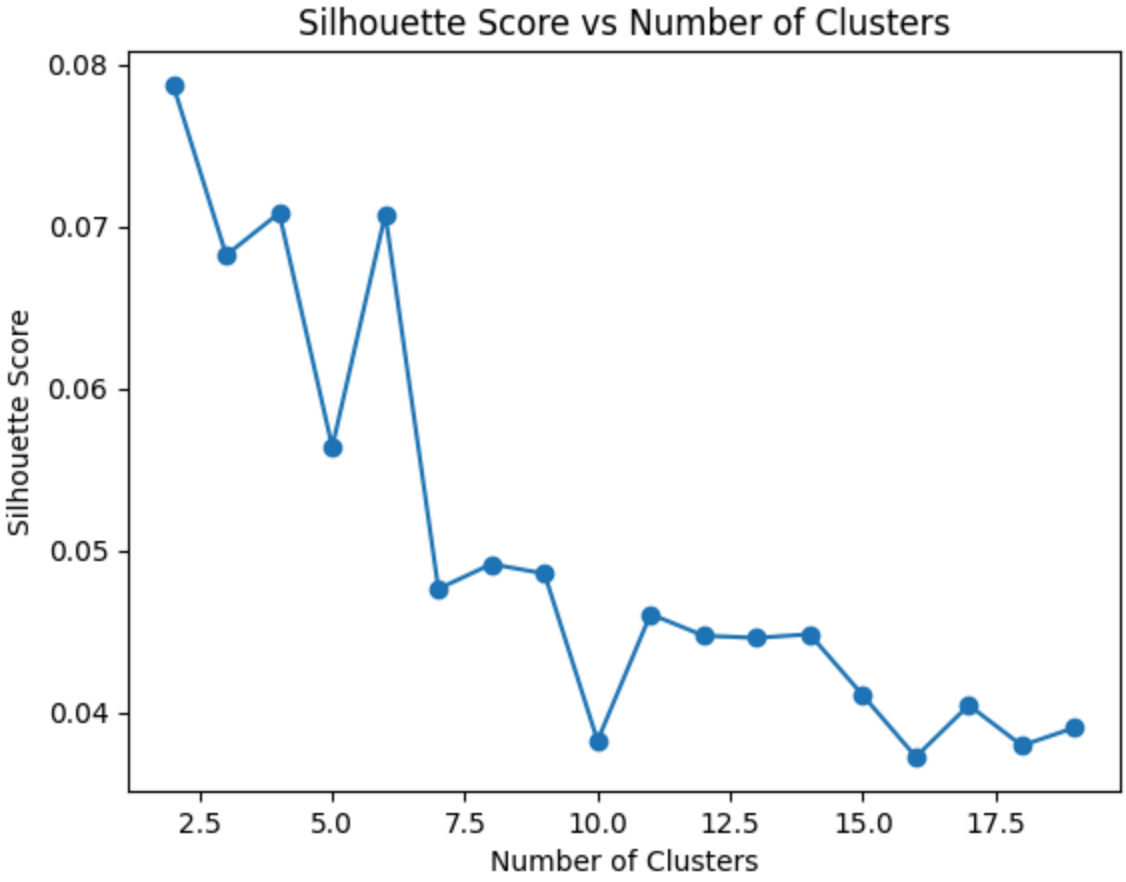
In [5]:
```
# Standardize the data
scaler = StandardScaler()
scaled_data = scaler.fit_transform(df)
```

3. Create a plot of the cluster silhouette score versus the number of clusters in a K-means cluster.

In [6]:
```
# Determine the optimal number of clusters using the silhouette score
silhouette_scores = []
cluster_range = range(2, 20)

# Evaluate silhouette score for each number of clusters
for n_clusters in cluster_range:
    kmeans = KMeans(n_clusters=n_clusters, random_state=42)
    cluster_labels = kmeans.fit_predict(scaled_data)
    score = silhouette_score(scaled_data, cluster_labels)
    silhouette_scores.append(score)

# Plot silhouette scores
plt.plot(cluster_range, silhouette_scores, marker='o')
plt.xlabel('Number of Clusters')
plt.ylabel('Silhouette Score')
plt.title('Silhouette Score vs Number of Clusters')
plt.show()
```




We can see that the largest silhouette score occurs at 2 clusters.

4. Use the plot created in (3) to choose an optimal number of clusters for K-means. Justify your choice.

In [7]:
```
# Get the max silhouette score and corresponding number of clusters
optimal_n_clusters = cluster_range[silhouette_scores.index(max(silhouette_sc
print(f"Optimal number of clusters chosen: {optimal_n_clusters}")
```

```
Optimal number of clusters chosen: 2
```

5. Fit a K-means model to the data with the optimal number of clusters chosen in part (4).

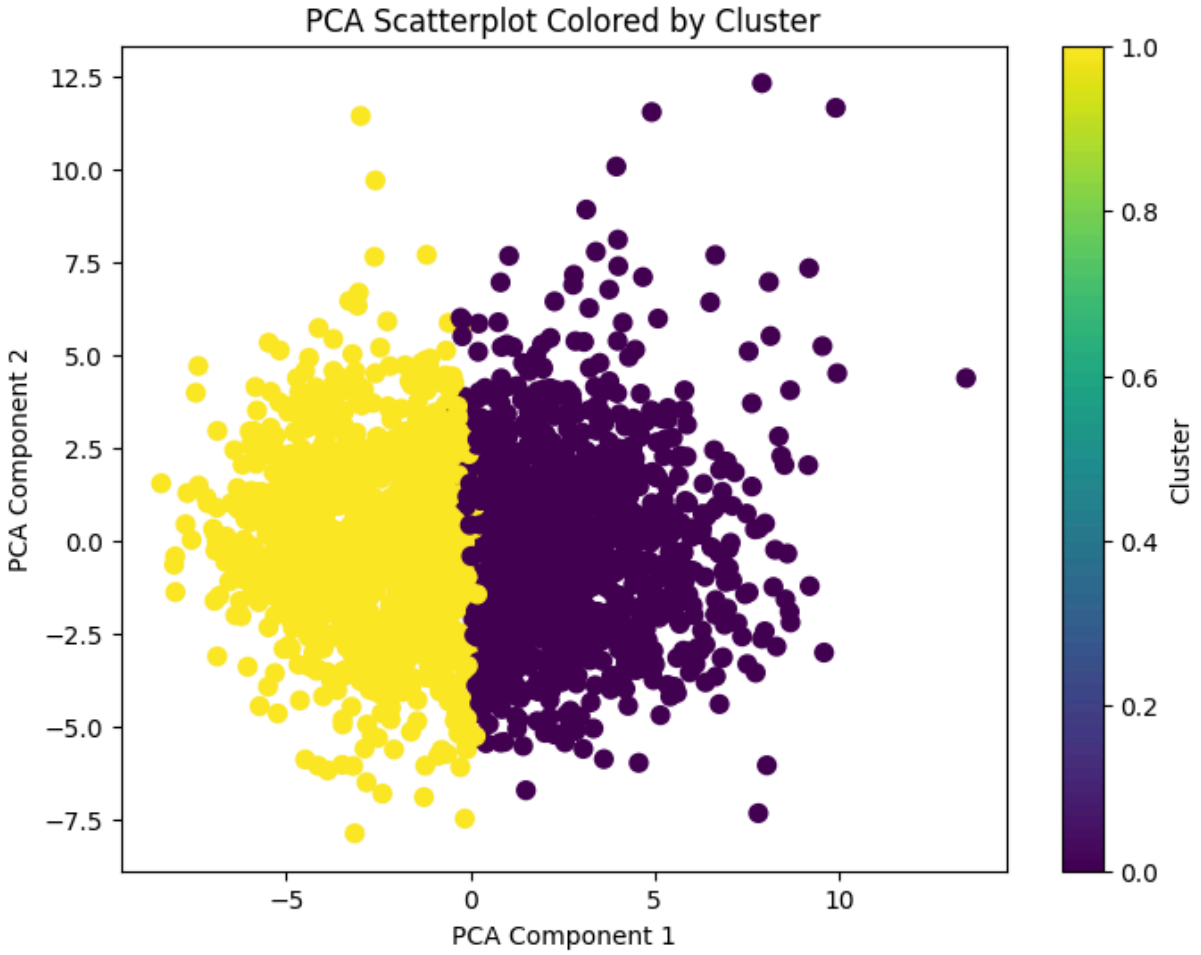
In [8]:
```
# Perform KMeans clustering with the optimal number of clusters
# As a reminder, KMeans is a clustering algorithm that partitions data into
kmeans = KMeans(n_clusters=optimal_n_clusters, random_state=42)
cluster_labels = kmeans.fit_predict(scaled_data)
```

6. Fit a PCA transformation with two features to the scaled data.

In [9]:
```
# PCA or Principal Component Analysis is a dimensionality reduction techniqu
pca = PCA(n_components=2)
pca_data = pca.fit_transform(scaled_data)
```

7. Make a scatterplot the PCA transformed data coloring each point by its cluster value.

In [10]:
```
# Plot PCA results with clusters
plt.figure(figsize=(8, 6))
plt.scatter(pca_data[:, 0], pca_data[:, 1], c=cluster_labels, cmap='viridis'
plt.xlabel('PCA Component 1')
plt.ylabel('PCA Component 2')
plt.title('PCA Scatterplot Colored by Cluster')
plt.colorbar(label='Cluster')
plt.show()
```




8. Summarize your results and make a conclusion.

The clustering analysis identified an optimal number of clusters using the silhouette score, indicating distinct groupings within the ALS patient data. After scaling and dimensionality reduction with PCA, the scatterplot showed clear separation between clusters, suggesting meaningful patterns. These clusters may represent different ALS

progression profiles or subgroups, which could be valuable for further clinical investigation. Overall, the approach demonstrates that unsupervised learning can reveal structure in ALS patient data, potentially aiding in personalized treatment strategies.

We're going to try dropping more columns to see if it improves clustering.

In [13]:
```
drop_cols = [
    "Age_mean",
    "Gender_mean",
    "bp_diastolic_max",
    "bp_diastolic_median",
    "bp_diastolic_min",
    "bp_diastolic_range",
    "bp_systolic_max",
    "bp_systolic_median",
    "bp_systolic_min",
    "bp_systolic_range",

] # This also set optimal number to 2 but the silo score was more much highe

# Compare clustering performance between original and cleaned data
df_original = df  # Original with basic drops
df_cleaned = df.drop(drop_cols , axis=1)  # Cleaned with additional drops

# Scale both datasets
scaler_orig = StandardScaler()
scaled_data_orig = scaler_orig.fit_transform(df_original)

scaler_clean = StandardScaler()
scaled_data_clean = scaler_clean.fit_transform(df_cleaned)

# Calculate silhouette scores for both datasets
silhouette_scores_orig = []
silhouette_scores_clean = []
cluster_range = range(2, 15)

for n_clusters in cluster_range:
    # Original data
    kmeans_orig = KMeans(n_clusters=n_clusters, random_state=42)
    cluster_labels_orig = kmeans_orig.fit_predict(scaled_data_orig)
    score_orig = silhouette_score(scaled_data_orig, cluster_labels_orig)
    silhouette_scores_orig.append(score_orig)

    # Cleaned data
    kmeans_clean = KMeans(n_clusters=n_clusters, random_state=42)
    cluster_labels_clean = kmeans_clean.fit_predict(scaled_data_clean)
    score_clean = silhouette_score(scaled_data_clean, cluster_labels_clean)
    silhouette_scores_clean.append(score_clean)

# Plot both silhouette scores on the same chart
plt.figure(figsize=(10, 6))
plt.plot(cluster_range, silhouette_scores_orig, marker='o', label='Original
plt.plot(cluster_range, silhouette_scores_clean, marker='s', label='Cleaned
plt.xlabel('Number of Clusters')
```

```
plt.ylabel('Silhouette Score')
plt.title('Silhouette Score Comparison: Original vs Cleaned Data')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()

# Print max scores for comparison
max_orig = max(silhouette_scores_orig)
max_clean = max(silhouette_scores_clean)
print(f"Max silhouette score (Original): {max_orig:.3f}")
print(f"Max silhouette score (Cleaned): {max_clean:.3f}")
print(f"Improvement: {max_clean - max_orig:.3f}")
```




```
Max silhouette score (Original): 0.079
Max silhouette score (Cleaned): 0.087
Improvement: 0.008
```